ΔV Efficiency Comparison of Hohmann and Bielliptic Orbital Transfers

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This report presents a comparative analysis of the Hohmann transfer and the Bielliptic transfer for transferring a satellite from a smaller circular orbit to a larger final orbit. The method of comparison used has normalized all values to the radius of the initial circular orbit, so that the method can be generally applied to any given system. Numerical examples are used to demonstrate the process of applying the method described in the report. It's found the Hohmann transfer is more efficient for $r_C/r_A < 11.94$, the Bielliptic transfer is more efficient for $r_C/r_A > 15.58$, but there's ambiguity for $11.94 < r_C/r_A < 15.58$. Lastly, the difference between the Hohmann and Bielliptic efficiency decreases as r_B/r_A increases.

I. Introduction

Orbital transfer maneuvers are fundamental to satellite mission design, and an important factor of the mission planning phase of any mission. When a satellite must transfer from a smaller circular orbit to a larger circular orbit, there are two types of maneuvers that have been determined to be the most efficient. These are the Hohmann transfer and the Bielliptic, which are two of the most studied orbital maneuvers in the field of astrodynamics.

The Hohmann transfer, created by Walter Hohmann in 1925, uses a single elliptical transfer orbit to travel from the initial circular orbit to the final circular orbit. When a satellite is in the initial circular orbit, an impulse is applied to change from it's initial orbit to the elliptical transfer orbit. The satellite then travels 180° from the transfer orbit's periapsis to apoapsis. Lastly, a second impulse is applied at apoapsis to change from the elliptical transfer orbit to the final circular orbit. The Hohmann transfer has been found to provide the lowest ΔV in many, but not all, transfer scenarios.

The Bielliptic transfer builds on the ideas of the Hohmann transfer, but adds a second elliptic transfer orbit to reduce the ΔV in wide range of transfer scenarios in which the Hohmann transfer is less efficient. A satellite starts in it's initial circular orbit and applies an impulse to shift to the first elliptical transfer orbit. The satellite then travels 180° from periapsis to apoapsis of the first transfer orbit. At this point, a second impulse is applied to shift to the second elliptical transfer orbit. The satellite then travels 180° from apoapsis to periapsis of the second transfer orbit. Lastly, a third impulse is applied to shift into the final circular orbit.

The goal of this project is to quantify and visualize the conditions under which each transfer is more fuel-efficient. Specifically, this report:

1) Derives the ΔV formulas for both transfer types
2) Implements MATLAB codes to generate comparison figures including a separatrix curve
3) Plots transfer trajectories for representative mission scenarios
4) Analyzes the turning points of the bi-elliptic ΔV curves relative to the Hohmann baseline

II. Theoretical Background

A. Two-Body Problem and Vis-Viva Equation

All maneuvers in this study are analyzed within the framework of the two-body problem, in which a satellite of negligible mass orbits a central body with no effects of surrounding bodies. The following equations are fundamental to the two-body problem and the concepts of this report. Given the specific angular momentum h, eccentricity e, and true anomaly θ the radial distance r of the satellite can be found at point using the trajectory equation from Curtis [1]:

$$r = \frac{h^2}{\mu}\frac{1}{1 + e\cos\theta} \tag{1}$$

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Given the semi-major axis a and the radial distance r at any point of an orbit, the velocity can be found by the Vis-Viva equation from Curtis [1]:

$$v = \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)} \tag{2}$$

For a circular orbit, the equation can be simplified because the semi-major axis a is equal to the radial distance r:

$$v_c = \sqrt{\frac{\mu}{r}} \tag{3}$$

B. Hohmann Transfer

The Hohmann transfer moves a satellite from a circular orbit of radius r_A to a circular orbit of radius $r_C > r_A$ using two impulsive burns along a single transfer ellipse with semi-major axis:

$$a_H = \frac{r_A + r_C}{2} \tag{4}$$

The velocity on the transfer ellipse at periapsis ($r = r_A$) and apoapsis ($r = r_C$) is obtained from Eq. (2):

$$v_{H,A} = \sqrt{\mu\left(\frac{2}{r_A} - \frac{1}{a_H}\right)} \tag{5}$$

$$v_{H,C} = \sqrt{\mu\left(\frac{2}{r_C} - \frac{1}{a_H}\right)} \tag{6}$$

The two ΔV burns are:

$$\Delta v_1^H = v_{H,A} - v_c \tag{7}$$

$$\Delta v_2^H = v_c - v_{H,C} \tag{8}$$

Both burns are positive, and can be combined into a total Hohmann ΔV equation:

$$\Delta \bar{v}_H = \frac{\Delta v_1^H + \Delta v_2^H}{v_A} \tag{9}$$

where $\bar{(\cdot)}$ denotes normalization by the initial circular speed v_A.

C. Bi-Elliptic Transfer

The bi-elliptic transfer uses three impulsive burns and two elliptical transfer orbits to travel from r_A to r_C via an intermediate point $r_B > r_C$. Using Eq. (2):, the velocity at each point an impulse is applied can be found as follows:

First transfer ellipse:

$$v_{1,A} = \sqrt{\mu\left(\frac{2}{r_A} - \frac{1}{a_1}\right)} \tag{10}$$

$$v_{1,B} = \sqrt{\mu\left(\frac{2}{r_B} - \frac{1}{a_1}\right)} \tag{11}$$

Second transfer ellipse:

$$v_{2,B} = \sqrt{\mu\left(\frac{2}{r_B} - \frac{1}{a_2}\right)} \tag{12}$$

$$v_{2,C} = \sqrt{\mu\left(\frac{2}{r_C} - \frac{1}{a_2}\right)} \tag{13}$$

The equations can be used to find the ΔV at each impulse:

$$\Delta v_1^{BE} = v_{1,A} - v_A \tag{14}$$

$$\Delta v_2^{BE} = v_{2,B} - v_{1,B} \tag{15}$$

$$\Delta v_3^{BE} = v_C - v_{2,C} \tag{16}$$

Note that Δv_1^{BE} and Δv_2^{BE} are both positive burns and Δv_3^{BE} is a negative burn the for $r_B > r_C > r_A$. The total normalized bi-elliptic ΔV is:

$$\Delta \bar{v}_{BE} = \frac{|\Delta v_1^{BE}| + |\Delta v_2^{BE}| + |\Delta v_3^{BE}|}{v_c(r_A)} \tag{17}$$

D. Comparison and the Separatrix

By working in normalized units (setting $r_A = 1$, so $v_A = 1$), all ΔV expressions depend only on the dimensionless ratios $\alpha = r_C/r_A$ and $\beta = r_B/r_A$. It's necessary that $\beta > \alpha$ for the following equations and examples. The two critical values of $\alpha = r_C/r_A$ are:

- $\alpha = r_C/r_A = 11.94$: Below this value, the Hohmann transfer is always more efficient regardless of r_B.
- $\alpha = r_C/r_A = 15.58$: Above this value, the bi-elliptic transfer is always more efficient for $r_B \to \infty$.
- $\alpha = 11.94 < r_C/r_A < 15.58$: Between these values, the preferred transfer depends on r_B.

Substituting Eq. (2) into (9) and Eq. (17) and using α and β as the function variables yields the following crucial equations from Curtis [1]:

$$\Delta \bar{v}_H = \frac{1}{\sqrt{\alpha}} - \frac{\sqrt{2}\,(1-\alpha)}{\sqrt{\alpha\,(1+\alpha)}} - 1 \tag{18}$$

$$\Delta \bar{v}_{BE} = \left(\sqrt{\frac{2\beta}{1+\beta}} - 1\right) + \left(\sqrt{\frac{2\alpha}{\beta(\alpha+\beta)}} - \sqrt{\frac{2}{\beta(1+\beta)}}\right) + \left(\sqrt{\frac{2\beta}{\alpha(\alpha+\beta)}} - \frac{1}{\sqrt{\alpha}}\right) \tag{19}$$

These equations are used as the primary calculation method for the numerical results. They are graphically represented in the following figures:

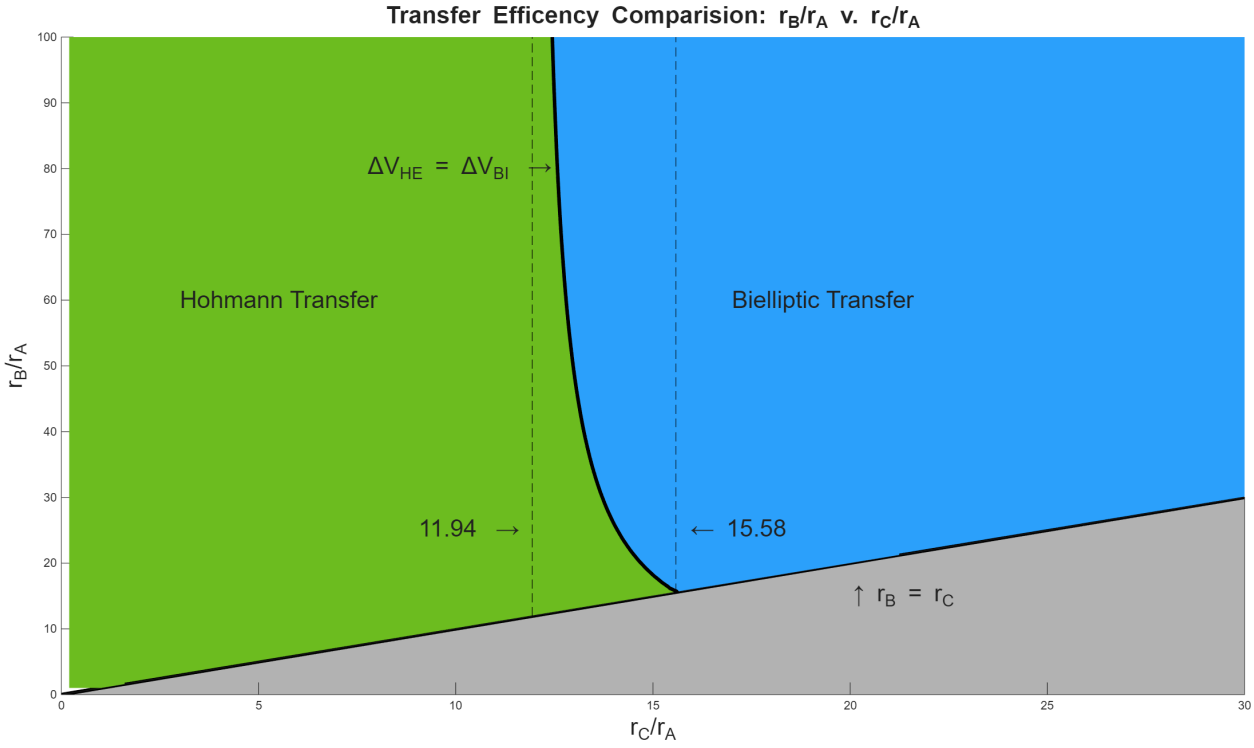


Fig. 1 Eq. (18) and Eq. (19) represented on a plot with α on the x-axis and β on the y-axis.

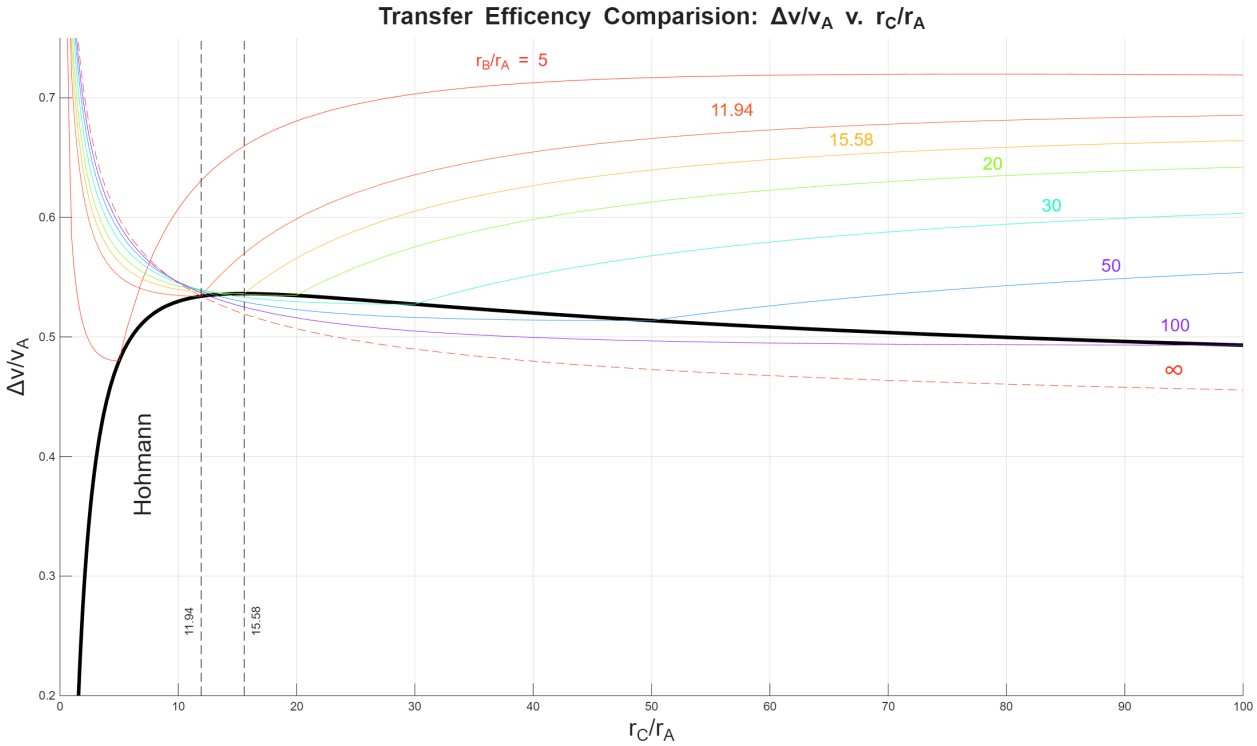


Fig. 2 Eq. (18) and various values of Eq. (19) plotted with α on the x-axis and $\Delta v/v_A$ on the y-axis.

III. Numerical Results

A. Numerical Examples using Fig 1.

Three coordinates were selected based off Fig. 1, one coordinate from each region of the plot:

- $(\alpha, \beta) = (10, 40)$ — bi-elliptic more efficient region
- $(\alpha, \beta) = (25, 40)$ — bi-elliptic more efficient region
- $(\alpha, \beta) = (25, 20)$ — Hohmann more efficient region

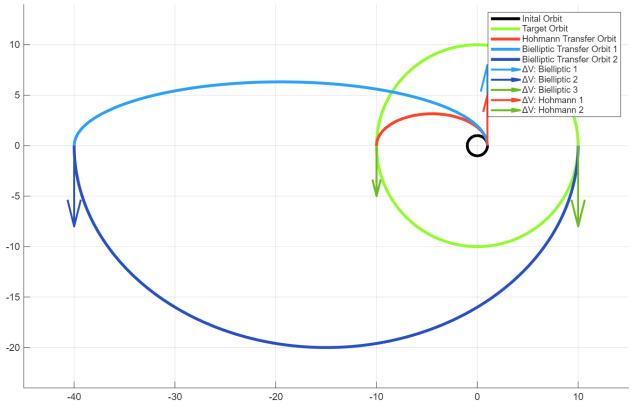


Fig. 3 Transfer trajectories for $(\alpha, \beta) = (10, 40)$

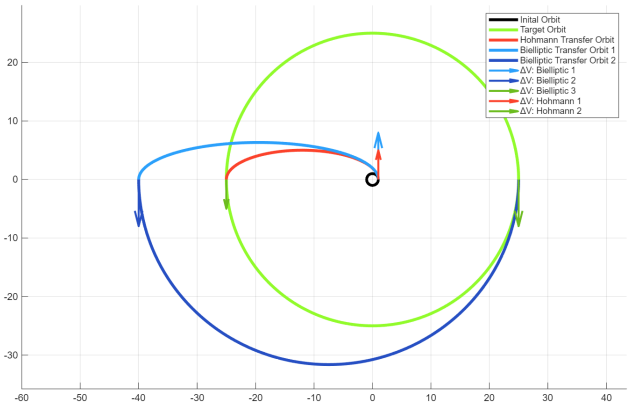


Fig. 4 Transfer trajectories for $(\alpha, \beta) = (25, 40)$.

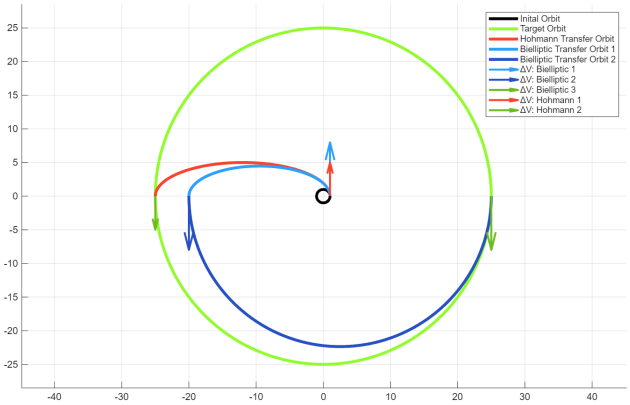


Fig. 5 Transfer trajectories for $(\alpha, \beta) = (25, 20)$.

Table 1 ΔV Results

r_C/r_A	r_B/r_A	$\Delta\bar{v}_{BE}$	$\Delta\bar{v}_H$
10	40	0.5298	0.5457
25	40	0.5313	0.5225
25	20	0.5313	0.5354

For the coordinate $(\alpha, \beta) = (10, 40)$, the Hohmann transfer requires a smaller Δv, while for $(\alpha, \beta) = (25, 40)$ the Bielliptic transfer requires a smaller Δv. These results reinforce the seperatrix curve results that a $\alpha < 11.94$ will always result in the Hohmann transfer being more efficient, and a $\alpha > 15.58$ will always result in the Bielliptic transfer being more efficient. The coordinates $(\alpha, \beta) = (25, 20)$ is an example of coordinates that are undefined for the Bielliptic transfer because it violates the condition $\alpha > \beta$.

B. Separatrix Curve

The separatrix curve shown in Fig. 1 is the locus of points (α, β) satisfying the equation:

$$\Delta\bar{v}_{BE}(\alpha, \beta) - \Delta\bar{v}_H(\alpha) = 0 \tag{20}$$

Algorithm

The separatrix curve was found using an implicit contouring approach with a two-dimensional parameter grid. A uniform 500×500 grid was constructed over the parameter space $\alpha = r_C/r_A \in [0, 100]$ and $\beta = r_B/r_A \in [1.01, 100]$. At each grid point, the ratio function

$$Z = \frac{\Delta\bar{v}_H}{\Delta\bar{v}_{BE}} - 1 \tag{21}$$

was evaluated, which equals zero at the separatrix condition Eq. (20). Grid points violating the physical constraint $\beta > \alpha$ were removed before plotting. MATLAB's `contourf` function was then used to extract the zero contour of Z by interpolating between adjacent grid points where Z changes sign, yielding the separatrix.

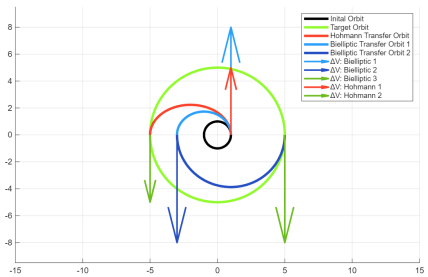
Challenges and Computational Considerations

The implicit contour approach proved to be effective in providing accurate results while being computationally efficient. The computational time for the chosen grid size was below 2 seconds. Compared to a root-solving approach, this method requires much less computing time. A key consideration of the implicit contour method is the resolution of the grid. If the grid is too coarse, it introduces interpolation error in the separatrix location, particularly near the asymptotic boundaries at 11.94 and 15.58. A grid of 500×500 points was found to provide good accuracy and remain very computationally efficient.

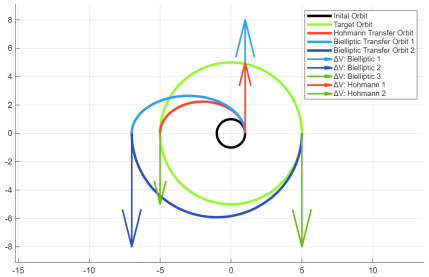
C. Turning Points on the $\Delta V/v_A$ Curves

In Fig. 2, curves of $\Delta\bar{v}_{BE}$ vs. r_C/r_A are plotted for various values. The values $\beta \in \{5, 15.58, 30\}$ were selected for understanding Fig. 2. Each curve exhibits a turning (local minimum) point that coincides with the Hohmann transfer curve at the separatrix condition, Eq. (20).

For each of the three r_B/r_A curves, two representative points were selected—one below and one above the turning point—and the corresponding transfer trajectories were plotted. The six trajectory plots are shown in the following figures.

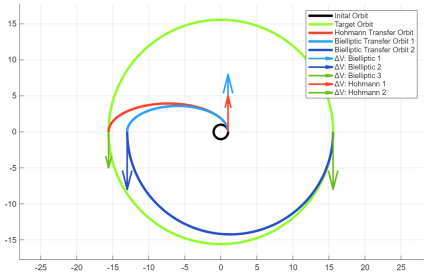


(a) Transfer trajectories for $(\beta, \alpha) = (5, 3)$.

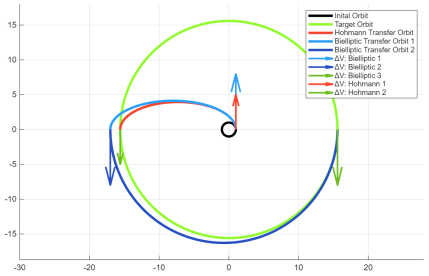


(b) Transfer trajectories for $(\beta, \alpha) = (5, 7)$.

Fig. 6 $\beta = 5$

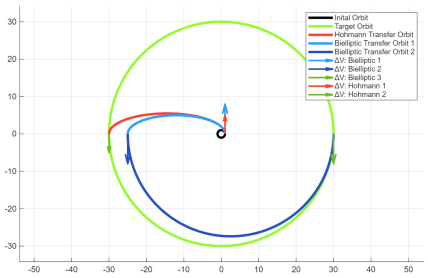


(a) Transfer trajectories for $(\beta, \alpha) = (15.58, 13)$.

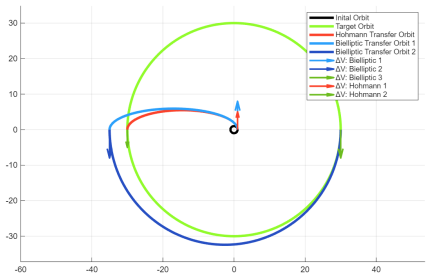


(b) Transfer trajectories for $(\beta, \alpha) = (15.58, 17)$.

Fig. 7 $\beta = 15.58$



(a) Transfer trajectories for $(\beta, \alpha) = (30, 25)$**.**



(b) Transfer trajectories for $(\beta, \alpha) = (30, 35)$**.**

Fig. 8 $\beta = 30$

Table 2 Δv **Results**

r_B/r_A	$\Delta\bar{v}_H$	r_C/r_A 1	$\Delta\bar{v}_{BE}$ 1	r_C/r_A 2	$\Delta\bar{v}_{BE}$ 2
5	0.4800	3	0.4021	7	0.5148
15.58	0.5363	13	0.5358	17	0.5362
30	0.5274	25	0.5317	35	0.5329

As demonstrated by the results, as the Bielliptic Δv approaches the turning-point, it gets close to the Hohmann Δv. Just like Table 1, the smaller $r_B/r_A 1$ values ae undefined for the Bielliptic transfer, but they still demonstrate the same concept of approaching the Hohmann Δv. As shown in Fig. 2, the turning point shifts to a larger value of α as β increases. There is also a smaller difference between the Hohmann and Bielliptic Δv's as β increases.

IV. Conclusions

This study compared the Hohmann and the Bielliptic orbital transfers using MATLAB-generated comparison figures and trajectory plots. Numerical examples were used to support these figures and plots. The following conclusions were drawn:

1) The Hohmann transfer is more efficient for $r_C/r_A < 11.94$, confirmed by all computed Δv values in that region.
2) For $r_C/r_A > 15.58$, the Bielliptic transfer with any finite $r_B > r_C$ is more efficient than the Hohmann transfer.
3) Between $11.94 < r_C/r_A < 15.58$, efficiency depends on the chosen r_B/r_A; a larger r_B favors the Bielliptic strategy.
4) The Implicit Contour approach to calculating the separatrix curve for the $11.94 < r_C/r_A < 15.58$ region provides a computationally efficient way to accurately calculate where the Δv for the Hohmann and Bielliptic transfers are equal.
5) The turning points on the $\Delta\bar{v}_{BE}$ curves represent the exact crossover condition where the Bielliptic and Hohmann Δv costs are equal.
6) The difference between the Hohmann and Bielliptic Δv's decreases as β increases.

References

[1] Curtis, H. D., *Orbital Mechanics for Engineering Students*, 4th ed., Revised Reprint, Butterworth-Heinemann, Elsevier, Oxford, 2021.

Appendix

Listing 1 Figure 1 Code

```
%%% ——— Transfer Efficiency Graph 1 ——— %%%
```

```
clear; clc, close all

% Creates and formats the graph
figure();
hold on
xlim([0 30])
ylim([0 100])
xlabel("r_C/r_A",'FontSize',20)
ylabel("r_B/r_A",'FontSize',20)
title('Transfer Efficency Comparision: r_B/r_A v. r_C/r_A','FontSize',20)

% Creates bound for B > A in graph
bound = @(a) a;
fplot(bound,[0 30],'k','LineWidth',3)          % Plots line
fill([0 30 30],[0 0 30],'k','FaceAlpha',0.3)   % Fills in grey below the line

% Creates line defining where both transfers have the same deltaV
a = linspace(0,100,500);
b = linspace(1.01,100,500);
[A,B] = meshgrid(a,b);
Z = ((1./sqrt(A)) - (sqrt(2).*(1-A))./sqrt(A.*(1+A)) - 1)./(sqrt((2.*(A+B))./(A.*B)) - (1+
Z(B<=A) = NaN;                    % Removes values where b < a

contourf(A,B,Z,[-Inf 0 Inf],'k','LineWidth',3) % Plots line where the deltaV's are equal a
colormap([108 188 31; 43 160 251]/255);        % Defines more user-friendly colors

% Adds annotations to communicate what each region and line represents
text(3,60,'Hohmann Transfer','FontSize',20)
text(17,60,'Bielliptic Transfer','FontSize',20)
text(20,15,'\uparrow r_B = r_C','FontSize',20)
text(7.7,80,' V_H_E  =  V_B_I  \rightarrow','FontSize',20)

% Adds lines to define ambigious region between hohmann and bielliptic
% deltaV's
line([11.94 11.94],[11.94 100],'Color','k','LineStyle','--')
line([15.58 15.58],[15.58 100],'Color','k','LineStyle','--')
text(9,25,'11.94 \rightarrow','FontSize',20)
text(15.88,25,'\leftarrow 15.58','FontSize',20)
```

Listing 2 Figure 2 Code

```
%%% ------- Transfer Efficiency Graph 2 ------- %%%

% Creates and formats the graph
figure();
hold on
grid("on")
xlim([0 100]);
ylim([0.2 0.75]);
xlabel('r_C/r_A','FontSize',20)
ylabel(' v /v_A','FontSize',20)
title('Transfer Efficency Comparision:  v /v_A v. r_C/r_A','FontSize',20)

% Equation for Hohmann line
dv_ho_vA = @(a) 1/sqrt(a) - (sqrt(2)*(1-a))/sqrt(a*(1+a)) - 1;
```

```
% Reference equation for Bielliptic lines
% dv_bi_vA = @(a,b) (sqrt(2*b/(1+b))-1) + (sqrt(2*a/(b*(a+b))) - sqrt(2/(b*(1+b)))) + (sqr

% Creates Bielliptic equations for each b value
dv_bi_vA_5 = @(a) abs(sqrt(2*5/(1+5))-1) + abs(sqrt(2*a/(5*(a+5))) - sqrt(2/(5*(1+5)))) + a
dv_bi_vA_1194 = @(a) abs(sqrt(2*11.94/(1+11.94))-1) + abs(sqrt(2*a/(11.94*(a+11.94))) - sq
dv_bi_vA_1558 = @(a) abs(sqrt(2*15.58/(1+15.58))-1) + abs(sqrt(2*a/(15.58*(a+15.58))) - sq
dv_bi_vA_20 = @(a) abs(sqrt(2*20/(1+20))-1) + abs(sqrt(2*a/(20*(a+20))) - sqrt(2/(20*(1+20
dv_bi_vA_30 = @(a) abs(sqrt(2*30/(1+30))-1) + abs(sqrt(2*a/(30*(a+30))) - sqrt(2/(30*(1+30
dv_bi_vA_50 = @(a) abs(sqrt(2*50/(1+50))-1) + abs(sqrt(2*a/(50*(a+50))) - sqrt(2/(50*(1+50
dv_bi_vA_100 = @(a) abs(sqrt(2*100/(1+100))-1) + abs(sqrt(2*a/(100*(a+100))) - sqrt(2/(100
dv_bi_vA_inf = @(a) (sqrt(2)-1) + (sqrt(2)-1)*sqrt(1/a);

% Plots teh Hohmann equation and each Bielliptic equation
fplot(dv_ho_vA,'k','LineWidth',3)
fplot(dv_bi_vA_5,'Color','#fb422b')
fplot(dv_bi_vA_1194,'Color','#fb602b')
fplot(dv_bi_vA_1558,'Color','#fbbd2b')
fplot(dv_bi_vA_20,'Color','#91fb2b')
fplot(dv_bi_vA_30,'Color','#2bfbcf')
fplot(dv_bi_vA_50,'Color','#2ba0fb')
fplot(dv_bi_vA_100,'Color','#8d2bfb')
fplot(dv_bi_vA_inf,'Color','#fb422b','LineStyle','--')

% Plots lines to show ambigious region between hohmann and bielliptic
% deltaV's
line([11.94 11.94],[0.2 0.8],'Color','k','LineStyle','--')
line([15.58 15.58],[0.2 0.8],'Color','k','LineStyle','--')

% Text annotations to identify each line on the graph
fontsize = 15;
text(35,0.73,'r_B/r_A = 5','FontSize',fontsize,'Color','#fb422b')
text(55,0.69,'11.94','FontSize',fontsize,'Color','#fb602b')
text(65,0.665,'15.58','FontSize',fontsize,'Color','#fbbd2b')
text(78,0.645,'20','FontSize',fontsize,'Color','#91fb2b')
text(83,0.61,'30','FontSize',fontsize,'Color','#2bfbcf')
text(88,0.56,'50','FontSize',fontsize,'Color','#8d2bfb')
text(93,0.51,'100','FontSize',fontsize,'Color','#8d2bfb')
text(93,0.47,'\infty','FontSize',25,'Color','#fb422b')
text(7,0.35,'Hohmann','FontSize',20,'LineWidth',3,'Rotation',90)
text(10.94,0.25,'11.94','Rotation',90)
text(16.58,0.25,'15.58','Rotation',90)
```

Listing 3 Trajectory Calculation and Plot Code

```
%%% ======= Transfer Efficiency Example 1 ======= %%%

clear; clc; close all

dv = struct();

% Given orbit radii ratios
[dv.ho1, dv.bi1] = transfer(10,40);
[dv.ho2, dv.bi2] = transfer(25,40);
```

```
[dv.ho3, dv.bi3] = transfer(25,20);

[dv.ho4, dv.bi4] = transfer(5,3);
[dv.ho5, dv.bi5] = transfer(5,7);
[dv.ho6, dv.bi6] = transfer(15.58,13);
[dv.ho7, dv.bi7] = transfer(15.58,17);
[dv.ho8, dv.bi8] = transfer(30,25);
[dv.ho9, dv.bi9] = transfer(30,35);

% Function to create plots and determine efficency
function [dv_ho, dv_bi] = transfer(a,b)
    figure;
    hold on

    % Delta V Calcualtions
    dv_ho = 1/sqrt(a) - (sqrt(2)*(1-a))/sqrt(a*(1+a)) - 1;
    dv_bi = (sqrt(2*b/(1+b))-1) + (sqrt(2*a/(b*(a+b))) - sqrt(2/(b*(1+b)))) + (sqrt(2*b/(a

    % Orbit Setup
    rA = 1;
    rB = b.*rA;
    rC = a.*rA;

    % Circular Orbits (Initial & Final Orbits)
    th_cir = linspace(0,2*pi,300);
    x1 = rA.*cos(th_cir);
    y1 = rA.*sin(th_cir);
    x2 = rC.*cos(th_cir);
    y2 = rC.*sin(th_cir);

    plot(x1,y1,'k','LineWidth',3,'DisplayName','Inital Orbit')
    plot(x2,y2,'Color','#91fb2b','LineWidth',3,'DisplayName','Target Orbit')

    % Hohmann Orbit (Transfer Orbit)
    th_el = linspace(0,pi,300);
    at = (rA + rC)./2;
    et = (rC - rA)./(rC + rA);
    rt = at.*(1 - et.^2)./(1 + et.*cos(th_el));
    xt = rt.*cos(th_el);
    yt = rt.*sin(th_el);

    plot(xt,yt,'Color','#fb422b','LineWidth',3,'DisplayName','Hohmann Transfer Orbit')

    % Bielliptic Orbits (Transfer Orbits)
    th_el1 = linspace(0,pi,300);
    th_el2 = linspace(pi,2*pi,300);

    at1 = (rA + rB)./2;
    at2 = (rB + rC)./2;

    et1 = (rB - rA)./(rB + rA);
    et2 = (rB - rC)./(rC + rB);

    rt1 = at1.*(1 - et1.^2)./(1 + et1.*cos(th_el1));
```

```
    rt2 = at2.*(1 - et2.^2)./(1 + et2.*cos(th_el2));

    xt1 = rt1.*cos(th_el1);
    yt1 = rt1.*sin(th_el1);
    xt2 = rt2.*cos(th_el2);
    yt2 = rt2.*sin(th_el2);

    plot(xt1,yt1,'Color','#2ba0fb','LineWidth',3,'DisplayName','Bielliptic Transfer Orbit
    plot(xt2,yt2,'Color','#264fc3','LineWidth',3,'DisplayName','Bielliptic Transfer Orbit

    % Plot Annotations
    legend;
    axis equal padded
    grid on

    % Bielliptic deltaV arrows
    quiver(rA,0,0,1,8,'Color','#2ba0fb','DisplayName',' V : Bielliptic 1','LineWidth',2,'M
    quiver(-rB,0,0,-1,8,'Color','#264fc3','DisplayName',' V : Bielliptic 2','LineWidth',2,
    quiver(rC,0,0,-1,8,'Color','#6cbc1f','DisplayName',' V : Bielliptic 3','LineWidth',2,'

    % Hohmann deltaV arrows
    quiver(rA,0,0,1,5,'Color','#fb422b','DisplayName',' V : Hohmann 1','LineWidth',2,'MaxH
    quiver(-rC,0,0,-1,5,'Color','#6cbc1f','DisplayName',' V : Hohmann 2','LineWidth',2,'Ma
end
```